

ECM3412

UNIVERSITY OF EXETER

**COLLEGE OF ENGINEERING, MATHEMATICS
AND PHYSICAL SCIENCES**

COMPUTER SCIENCE

Examination, May 2017

Nature-Inspired Computation

Module Leader: Professor Edward Keedwell

Duration: TWO HOURS

Answer question 1, and two out of the other three questions.
Question 1 is worth 40 marks. Other questions are worth 30 marks each.

The marks for this module are calculated from 70% of the percentage mark for this paper plus 30% of the percentage mark for associated coursework.

No electronic calculators of any sort are to be used during the course of this examination.

This is a CLOSED BOOK examination.

SECTION A (Compulsory Question)

Question 1

- (a) Write pseudocode for *particle swarm optimisation* (PSO). In your answer, you should include feasible parameter settings for each of the parameters associated with the algorithm.

(8 marks)

- (b) *Exhaustive search* is an algorithm that might be used as an alternative to a nature-inspired approach. Explain the benefits and drawbacks of using exhaustive search for optimisation problems.

(5 marks)

- (c) Evolutionary algorithms can make use of a number of different representation types to solve difficult search and optimisation problems.

- (i) Explain what is meant by a *direct representation* and an *indirect representation* in evolutionary algorithms, providing an example for each.

(5 marks)

- (ii) Give one advantage and one disadvantage of using indirect representations in evolutionary algorithms and explain why it is considered an advantage or disadvantage.

(4 marks)

- (d) Briefly describe the following symbols and their purpose in *ant colony optimisation*.

- (i) ρ

(2 marks)

- (ii) α

(2 marks)

- (iii) β

(2 marks)

- (e) Explain the following terms as they relate to cellular automata, illustrating your answer with examples where necessary.

- (i) Localism

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(2 marks)

(ii) Parallelism

(2 marks)

(iii) Homogeneity

(2 marks)

(f) During learning, *perceptrons* change their weights to reduce the error between the actual output and target output. With the aid of equations, explain how this is achieved for a perceptron using:

(i) The standard perceptron learning rule.

(ii) As (i) above with a learning rate.

(iii) As (ii) above with the *Widrow-Hoff* learning rule.

(6 marks)

(Total 40 marks)

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SECTION B (Optional Questions)

Question 2

Neural networks are useful pattern recognition tools and can also be used to study human cognition.

- (a) Describe the neural network approach (including representation) and, briefly, the experimental findings in the paper "Sejnowski, T. J. and Rosenberg, C. R. (1987) 'Parallel networks that learn to pronounce English text' *Complex Systems*"

(15 marks)

- (b) *Self-organising maps* (also known as *Kohonen networks*) are neural networks that are able to learn structure from data using unsupervised learning.

- (i) Explain what is meant by the word *unsupervised* in this context and how *unsupervised learning* differs from *supervised learning*.

(5 marks)

- (ii) Show, with the aid of equations, how a self-organising map learns from an input pattern presented to it.

(5 marks)

- (iii) Two variables involved in the learning process in a self-organising map will change over time. Explain what these variables are, how they change, and the effect this has on the learning process.

(5 marks)

(Total 30 marks)

Question 3

Working for a data analysis company, you have been asked to develop a regression model to determine the *miles per gallon (mpg)* for a car, given data relating to its *weight (in kgs)*, *number of seats*, *gearbox (number of gears)*, *engine size (in cc)* and *aerodynamic coefficient (coefficient of drag (cd))*. You are provided with a dataset of 100 cars with these features and known mpg values with which to train your model and decide to use *genetic programming* to develop it.

- (a) Genetic programming requires five preparatory steps before an optimisation can begin. Name these steps and for each step, describe how you would approach the problem described above.

(15 marks)

- (b) For each of the following, draw a diagram of the tree that you would expect to see for the example shown above:

- (i) A random tree with maximum depth 4.
- (ii) A mutation operation on the tree you have drawn in (i).
- (iii) A crossover operation between the tree you have drawn in (i) and another random tree of your construction.

(9 marks)

- (c) After running your genetic programming system for some time you notice that your algorithm is experiencing *bloat*, where trees become very large during evolution.

- (i) Describe the behaviour you would expect to see from the system as a result of bloat.
- (ii) Explain what measures you could take to reduce the likelihood of bloat.

(6 marks)

(Total 30 marks)

Question 4

Evolutionary algorithms (EA), ant colony optimisation (ACO) and particle swarm optimisation (PSO) are all examples of population based search.

- (a) Describe the advantages that population-based search brings to search and optimisation tasks over methods that use a single point strategy.

(5 marks)

- (b) Describe the additional algorithmic steps that an EA must conduct due to the existence of a population.

(5 marks)

- (c) Explain how communication between population members differs between ACO and PSO.

(5 marks)

These three algorithms have different methods of varying the extent to which they exploit or explore during search and optimisation. For each algorithm, describe the parameters that are modified to achieve different exploitation and exploration capability and explain how varying these alters the behaviour of the algorithm.

(15 marks)

(Total 30 marks)